A.I.M.S. (AUTOMATED INTERVENTION AND MASTERY SYSTEM): A WEB-BASED EDUCATIONAL PLATFORM WITH AI-DRIVEN ASSESSMENT AND DYNAMIC REMEDIATION

A Capstone Project Presented to the

Faculty of the College of Computer Studies

Carlos Hilado Memorial State University

Alijis Campus, Bacolod City

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Information Technology

By

Mikhoel Runo B. Orbigoso

Patrick R.Sagum

Maika T. Zambra

January 2027

Chapter 1

INTRODUCTION

Background of the Study

Technology is becoming a part of education systems across the globe to aid in teaching, learning and evaluation. Educational technologies that are powered by Artificial Intelligence (AI) have demonstrated the potential to deliver personalized learning experiences, adaptive feedback, and automated assessment capabilities capable of assisting educators to respond to the needs of different learners and enhance their academic performance. Studies have shown that AI-based tools have the potential to make instruction more efficient, customize learning strategies to accommodate personalized learning processes, and facilitate informed decision-making in academic environments (Merino-Campos, 2025; Marcos, 2026).

The use of technology in basic education in the Philippine context has been increasing, and the COVID-19 pandemic has contributed to the adoption of flexible learning modalities. Nevertheless, age-old challenges of digital divide, uneven access to learning resources, poor infrastructure, and manual assistance with evaluation and module distribution continue to be a problem of many public schools (Bustillo & Aguilos, 2022; Balase & Paglinawan, 2025). Teachers tend to use non-standardization means like group discussions to communicate modules and have to score and mark student performance manually, which deprives them of time to teach and provide individualized intervention. In addition, research on educational technology in the Philippines also states that although

digital tools such as LMS are employed, the lack of connectivity and digital skills hinders their maximum functionality (Colegado, 2025).

Those difficulties leave the necessity to develop more intelligent and systematic solutions that can be used to centralize content delivery, automate assessment procedures, and facilitate personalized intervention of learners. This requirement is especially appropriate to AI-based systems, which include automated question generation, dynamically-adaptive assessments, and analytics dashboards that can inform teaching and correction. The positive effect of such systems on engagement and learning outcomes is supported in the international literature in case such systems are implemented thoughtfully.

That being the case, the given study presents A.I.M.S. (Automated Intervention and Mastery System) an AI-powered web-based educational solution aimed at supporting the work of the public school teachers simplifying the work with modules and offering automated evaluation and dynamic remediation based on the performance of individual students. A.I.M.S. is expected to help decrease the administrative load, enhance the correctness of the tracking of performance, and assist in making timely decisions in instructions, by incorporating the use of features like Module Management, Automatic assessments, AI-powered quiz generation, automated grading, competency-based progression, and performance tracking into it. It is projected that this implementation of the A.I.M.S. platform will fill the recorded gaps in the use of technologies, increase the workflow of the teachers, and become stronger in terms of the results of the learners in governmental schools due to intelligent automation and the data-driven instructions.

Objective of the Study

The general objective of the study aims to design and develop a web-based application titled "A.I.M.S. (Automated Intervention and Mastery System): A Web-Based Educational Platform with AI-Driven Assessment and Dynamic Remediation".

Specifically, this aims the following:

1. design and develop a web-based application with the following technical features:
    1.1. centralized digital library for module distribution and management.
    1.2. ai-powered quiz generator with instructor validation controls.
    1.3. automated grading module with customizable rubrics.
    1.4. mastery-lock system for automated student remediation.
    1.5. student performance dashboard for real-time academic tracking.
2. test the functionality of the features mentioned above
3. evaluate the system using ISO 25010 standardize evaluation tool in terms of
    3.1. Functionality Suitability
    3.2. Performance Efficiency
    3.3. Usability
    3.4. Compatibility
    3.5. Reliability
    3.6. Security
    3.7. Maintainability
    3.8. Portability

Significance of the Study

A.I.M.S. (Automated Intervention and Mastery System): A Web-Based Educational Service with AI-Based Assessment and Dynamic Remediation is marketed to be of substantial and positive importance to the following groupings:

Teachers. Such system will help the teachers to have better control of learning modules, automation of assessment, production of adaptive questions and real time tracking of student performance. This will allow teachers to concentrate more on instruction delivery, individual support and intervention on students who require further guidance by lessening the administrative burdens, as well as, offering timely analytics.

Students. A.I.M.S. let students study learning modules and tests on the internet and get immediate feedback, as well as enjoy AI-based dynamic correction based on their personal performance. This promotes self-directed learning, enables learners to gain awareness of their areas of weakness and areas of strength and mastery of learning content.

Researchers. The research offers empirical evidence concerning the use of the AI-based web-based learning systems in the Philippine public schools environment. The effectiveness of the system to enhance the learning results, improve the process of assessments, and assist personalized learning can be studied by the researchers as the part of the overall literature on educational technology and AI-assisted learning.

Future Researchers. The results and development experience of A.I.M.S. can be used as a base in the future studies of intelligent learning platforms. Researchers may extend on the structure of the system in future, add features or investigate other AI

applications in education, creating new innovations to improve the teaching and learning experience in different teaching and learning environments.

Scope and Limitation of the Study

The paper is devoted to the design, development and testing of A.I.M.S. (Automated Intervention and Mastery System): A Web-Based Educational Platform with AI-Driven Assessment and Dynamic Remediation that is supposed to be used in the local school environment. Some of its core features covered by the system are module management which gives teachers the ability to post and distribute learning materials on-line or in printed form, automated evaluation which provides recording and scoring of student responses, automatic generation of assessment questions but with the ability of the teacher to control them, module distribution in a standardized form so that official learning material is made available to all students and automated grading with customizable rubrics to provide feedback and custom scoring. The system will be tested according to the ISO 25010 standard that determines functionality, performance, usability, compatibility, reliability, security, maintainability, and portability.

The study is limited in some ways in spite of its elaborate design. A.I.M.S. is mainly designed to serve the population schools and might have to be modified to suit the use in the privates or tertiary institutions. The online services of it require the stable connection of the internet, and the offline services are restricted with printing modules. The paper does not consider the long-term impacts on the academic performance of students because the analysis is on the system usability, functionality and short-term

effectiveness. Also, the automatic question generator is based on pre-operative algorithms, and it might be necessary to involve the teacher in order to meet the assessment requirements that are specific and contextual. These limitations stipulate the limits of the research as well as present a clear guideline towards assessing the functionality and effectiveness of the system in the target environment.

Definition of Terms

For a clearer understanding of the study, the following terms are defined both contextually based on existing literature and operationally as they are used in the context of A.I.M.S.

AI-Powered Quiz Generator. Contextually, this refers to the use of neural question generation techniques and natural language processing to automatically create educationally valuable assessment items from learning materials, ensuring both fluency and pedagogical relevance (Bulathwela et al., 2023).

Operationally, in this study, it refers to the module in A.I.M.S. that uses Google Gemini API to automatically generate multiple-choice, true/false, and short-answer quizzes aligned with DepEd competencies from teacher-uploaded documents.

Automated Grading Module. Contextually, it refers to the use of algorithms and natural language processing to evaluate student responses and assign scores based on predefined assessment criteria, reducing manual grading workload while maintaining consistency (Ramesh & Sanampudi, 2022).

Operationally, it is the system component in A.I.M.S. that evaluates quiz responses according to teacher-defined customizable rubrics, providing immediate automated score computation and instant feedback to students.

Centralized Digital Library. Contextually, this is an integrated online platform within a learning management system that organizes and distributes educational materials, enabling structured access to course content and resources (Bradley, 2021).

Operationally, it is the repository feature in A.I.M.S. that allows teachers to upload and organize learning modules by subject and grade level, enabling students to securely access, search, and download educational materials.

Mastery-Lock System. Contextually, in competency-based education, this is a progression mechanism where students acquire knowledge and complete formative assessments, proceeding to new content only after demonstrating mastery of prerequisite material through corrective pathways and retesting (Persky & Hughes, 2022).

Operationally, it is the conditional progression feature in A.I.M.S. that restricts access to subsequent lessons until students achieve 70 percent or higher on previous assessments, triggering automated remediation interventions for non-mastery.

Student Performance Dashboard. Contextually, it is a visual analytics interface that provides actionable insights into learning behaviors and academic progress through data-driven visualizations, enabling students to regulate their learning activities effectively (Susnjak et al., 2022).

Operationally, it is the analytics component in A.I.M.S. that displays interactive charts tracking student engagement metrics, quiz completion rates, mastery scores, and progression status for real-time academic monitoring.

ISO 25010. Contextually, it is the international software quality standard that defines eight characteristics for evaluating system and software products: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability (ISO/IEC, 2011).

Operationally, it is the standardized evaluation framework used by the researchers to rigorously assess A.I.M.S.' system quality, effectiveness, and user satisfaction through surveys and functional testing during the validation phase.